

MACReporting

Chapter Reporting & Data Management Application

PROJECT OVERVIEW

MACReporting is a web-based application designed to improve how chapter committee and post-event information is collected, stored, retrieved, and used. The current process captures valuable information, but using that information later can require manually searching individual reports. MACReporting is designed to move the chapter from **simply collecting reports to collecting usable data**.

THE PROBLEM

A document-based process works when someone needs to read one report. It becomes inefficient when leadership needs answers across many reports - especially for award submissions, annual reporting, historical review, attendance, program activity, expenses, outcomes, and accomplishments.

Finding those answers should not require opening reports one at a time.

PROPOSED SOLUTION

Authorized users will select a report type, create a new report or retrieve an existing report for update, and complete questions dynamically loaded from the selected template. Responses will be stored as **structured data** while the application also generates the formatted PDF needed for traditional reporting.

PDF Human-readable report	DATA Searchable and reusable	EXCEL Optional export
-------------------------------------	--	---------------------------------

HOW IT WILL WORK

- 1. Select Report Type** - Committee Report or Post-Mortem Report
- 2. Choose an Action** - Create New or Update Existing
- 3. Complete the Report** - Template-driven sections and questions load dynamically
- 4. Save the Data** - Responses are stored for future updates and analysis
- 5. Generate the Report** - A standardized PDF is created
- 6. Reuse the Data** - Search, query, or export authorized information

TECHNICAL APPROACH

Multi-tier web application with separate presentation, application/API, and data layers. Current proof of concept: **React** user interface, **Python/Flask** REST API and application logic, **SQLite** relational database, and **JSON** between the frontend and API. Production decisions will emphasize low operating cost, security, maintainability, data ownership, and portability.

MVP - PHASE 1

- Create Committee and Post-Mortem Reports
- Dynamically render template-driven questions
- Store responses as structured data
- Retrieve and update existing reports
- Generate PDFs and maintain report history
- Export reporting data to Excel
- Lock finalized reports against unauthorized updates

FUTURE DIRECTION - PHASE 2

Expand MACReporting into a **role-based reporting and analytics dashboard**. Authorized users could explore chapter data directly in the browser, filter by committee, event, report type, reporting period or date range, review summary metrics, drill into individual reports, search historical information, and export filtered data to Excel. Access would be role-based - for example, a Committee Chair could see committee-level information while authorized leadership could access chapter-wide data.

PROJECT GOAL: Information entered once should serve multiple purposes - chapter reporting, historical reference, analysis, dashboards, and award submissions - without repeatedly searching documents for information the chapter has already collected.